Digital Signal Processing Assignment # 12

1. A discrete-time LTI system has the following difference equation.

$$y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + \frac{1}{4}x(n) + \frac{1}{2}x(n-1) \tag{1}$$

   (a) Find the transfer function $H(z)$

   (b) Find the frequency response $H(\Omega)$, and plot with Matlab the amplitude and phase response.

   (c) Find the impulse response.

   (d) Plot the direct-form I, direct-form II, transposed direct-form II structure of the system.

   (e) Is the system stable?

2. A discrete-time LTI system has the impulse response $h(n) = [2, -1, 3, -1, 2]$.

   (a) Find the transfer function $H(z)$.

   (b) Find the frequency response $H(\Omega)$, and plot with Matlab the amplitude and phase response.

   (c) Find the difference equation representation.

   (d) Plot the direct-form structure of the system.

   (e) Is the system stable?

3. The transfer function of a LTI system is given as follows.

$$H(z) = \frac{1 - z^{-1} + \frac{1}{3}z^{-2}}{1 - \frac{1}{2}z^{-2}} \tag{2}$$

   (a) Find the difference equation.

   (b) Plot the direct-form II structure of the system.

   (c) Is the system stable?